

Physical Design Implementation Challenges in Highly Memory-Intensive Design in 40 nm

Memory-intensive blocks with vivid aspect-ratio memory lead to tight challenges in floor planning to placement in terms of congestion and timing. This article discusses a methodology developed to overcome challenges in implementing a flat design with 1.6 million instances and 640 macros. It also talks about specific timing, congestion and power related challenges from floor planning and power planning stages, and different placement techniques adopted to overcome congestion and timing

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Physical design implementation of a very large floor plan with high memory count poses multiple challenges mostly in areas like floor planning, power planning and also turnaround time for tool coupled with memory management. In addition to the huge memory count, variation in the size of memories multiplies the complexity and increases challenges in physical design implementation.

Here we talk about designing with a very high memory count of more than 600 and memories of different sizes with different pin densities, which brings along complexity of floor planning. Due to vivid sizes of memories, power planning is also equally challenging, and care has to be taken that rails are not being removed during this stage. With a large core and huge instance count, the run-times and memory requirements were very high, and this limited what the tool could be used for during flat implementation.

The block contained 21 main clocks and hence building the clock tree and meeting latencies and skew targets was also a challenge. We achieved these targets by using some customisation techniques for CTS.

Apart from this, the blocks had highly skewed floor plans, which

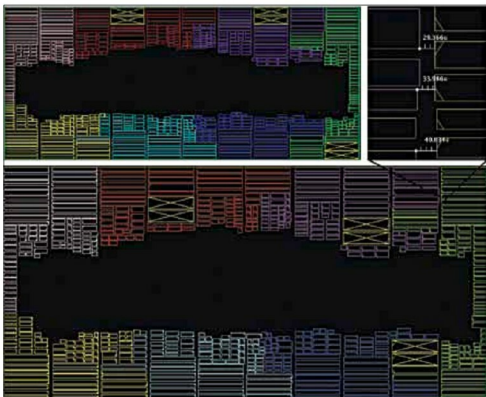


Fig. 1: Floor planning

don't lend themselves well to typical optimisation engines, and quite often, congestion and sub-optimal timing and placement complicated the implementations.

Design details

This block is over 12.699 mm², implemented in 40 nm, with a total instance count of 1.6 million and over 640 macros/memories. The pin count is 652 data and clock pins. The design

contains eight metal layers, of which six are used for block-level signal routing and the top two layers are used for power mesh and top-level routing.

The design utilisation is 63 per cent.

The block has 21 main clocks. The maximum frequency is 250 MHz and the minimum frequency is 50 MHz. The design has three modes and seven corners, two for setup and five for hold.